



NONLINEAR MODEL

UPA861TD

BJT NONLINEAR MODEL PARAMETERS⁽¹⁾

Parameters	Q1	Q2	Parameters	Q1	Q2
IS	8e-17	137e-18	MJC	0.53	0.24
BF	128	129	XCJC	1	0.3
NF	1	0.9992	CJS	0	0
VAF	17	22.4	VJS	0.75	0.75
IKF	0.18	2.8	MJS	0	0
ISE	3.3e-15	229e-15	FC	0.37	0.55
NE	1.48	2.5	TF	6e-12	5e-12
BR	9.05	81.7	XTF	11.9	0.05
NR	1.05	0.9944	VTF	9.55	0.5
VAR	4.3	1.9	ITF	1.78	0.005
IKR	0.009	0.018	PTF	69.1	0
ISC	4e-15	227e-18	TR	1e-9	1.0e-9
NC	2	1.17	EG	1.11	1.11
RE	0.8	0.75	XTB	0	0
RB	11.1	5	XTI	3	3
RBM	2.46	3	KF	0	117e-15
IRB	0.02	0.005	AF	1	1.34
RC	7.5	6			
CJE	0.415e-12	0.68e-12			
VJE	0.68	0.92			
MJE	0.53	0.26			
CJC	0.102e-12	0.16e-12			
VJC	0.29	0.64			

(1) Gummel-Poon Model

MODEL RANGE

Frequency: 0.1 to 3.0 GHz
Bias: $V_{CE} = 0.5\text{ V to } 2.5\text{ V}$, $I_C = 0.5\text{ mA to } 20\text{ mA}$
Date: 09/02



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